Lessons 8 Proof by Contradiction

Prove by contradiction.

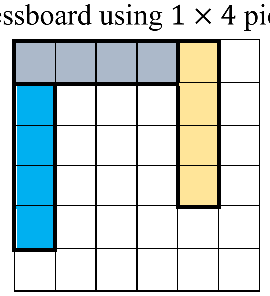
1. There is no largest even integer.
2. There is no largest irrational number.
3. There are no integers a and b such that $25a - 15b = 3$.
4. There are no integers x and y such that $4x^2 - y^2 = 1$.
   Hint Factorize $4x^2 - y^2$.
5. The only even prime is 2.
6. If the sum of two primes is prime, then one of the primes must be 2.
   Note: You may use the proposition in #5.
7. The sum of a rational and irrational number is irrational.
8. For all prime p, $\sqrt{p}$ is irrational. **Note**: You may use the following:
   For every prime p and all integers a, b, if $p|ab$, then $p|a$ or $p|b$.
9. $\sqrt[3]{2}$ is irrational.
10. Let S be a subset of $\mathbb{R}$ which is bounded below (see Exercise 4 in Lesson 5).
    We have an axiom called the Well-Ordering Principle (WOP):
    - Any nonempty subset of $\mathbb{Z}$ bounded below has a least element.

    This axiom is intuitive and "self-evident", and we accept it without proof.
    a) Find the least element of $A = \{x \in \mathbb{Z}: x \geq 0\}$ and $B = \{x \in \mathbb{Z}: x > 0\}$.
    However, if we replace $\mathbb{Z}$ in WOP by $\mathbb{R}$ or $\mathbb{Q}$, the statement is *no* longer true.
    b) Prove that the positive real numbers $\mathbb{R}^+ = (0, \infty)$ is bounded below, find the greatest lower bound (glb).
    c) Prove by contradiction that $\mathbb{R}^+$ has no smallest element.

For the following propositions, give a direct proof and a proof by contradiction.

11. If a and b are real numbers and $ab = 0$, then $a = 0$ or $b = 0$.
12. If a and b are real numbers and $a + b = 0$, then $a \leq 0$ or $b \leq 0$.
13. The product of two consecutive integers is non-negative.

(Optional) Prove by contradiction that it is not possible to cover a 6×6 chessboard using 1×4 pieces.



Hint Number the board as follows:

1	2	3	4	1	2
2	3	4	1	2	3
3	4	1	2	3	4
4	1	2	3	4	1
1	2	3	4	1	2
2	3	4	1	2	3

Every 1×4 piece covers "4" exactly once; how many 4's are on the board?